TOUHID IMAM

6745 W Coolidge St, Phoenix, AZ 85033 — (712) 577 8084
Touhid.Imam@coyotes.usd.edu
https://github.com/touhid-imam/data-analyst | linkedin.com/in/touhidimam/

Summary

Data Analyst with 5+ years of experience in data integration, analysis, and visualization across multiple industries, including healthcare. Proficient in SQL, Python, machine learning, and data visualization tools like Tableau and Power BI. Passionate about leveraging data to improve business outcomes and decision-making.

Experience

- **Intern Data Analyst**
  UpSkill (Remote)
  September 2025 – Present
  - Collaborated with cross-functional teams in the healthcare industry to collect, clean, and analyze complex medical data, transforming raw datasets into actionable business insights for improved patient care and operational efficiency.
  - Designed and implemented interactive dashboards and reports using Power BI and Tableau, providing real-time insights for healthcare professionals and decision-makers.
  - Utilized advanced analytical tools such as Python, SQL, and Excel to extract meaningful trends and patterns in healthcare data, supporting strategic business and clinical objectives.
  - Developed automated data processing workflows, ensuring seamless integration of healthcare data from various sources for analysis and reporting.
  - Applied Natural Language Processing (NLP) and data cleaning techniques to ensure the accuracy and usability of healthcare data.
  - Worked in an agile environment to deliver timely, data-driven solutions that align with healthcare organization goals and enhance service delivery.
- **Data Analyst**
  RockIT Fuel Design And Tech, Ontario, Canada
  November 2020 – August 2023
  - Processed and enriched client datasets using Python (Pandas, NumPy), SQL, and Excel to deliver reliable analytical insights.
  - Created Python automation scripts to optimize data workflows and enhance reporting efficiency.
  - Designed dynamic visualizations in Tableau and Power BI to reveal market trends, empowering client decision-making.
  - Executed SQL queries for customer behavior analysis and demand forecasting, strengthening strategic planning.
  - Developed data quality tools and automated checks, boosting dataset reliability and expediting project delivery in a collaborative setting.
- **Programmer and Data Analyst**
  Magic Technologies Group, California, USA
  April 2018 – October 2020

- Performed data manipulation, cleansing, and formatting of client datasets using Python (Pandas), SQL, and Excel to ensure high-quality inputs.
- Developed automated Python scripts to streamline data transformation and reporting, reducing processing time by 30%.
- Conducted exploratory data analysis and created visualizations with Power BI and Tableau, driving a 20% sales increase through customer segmentation.
- Wrote SQL queries to extract customer data for pricing insights and performance metrics, improving decision-making by 25%.
- Built internal tools and validation pipelines, enhancing data accuracy and accelerating delivery timelines by 30% in a collaborative environment.

Skills

- **Data Analysis & Processing:** SQL, Python (Pandas, NumPy, Scikit-learn), Excel, Data Wrangling, Feature Selection
- **Data Visualization:** Tableau, Power BI, Matplotlib, Seaborn, Excel Dashboards
- **Statistical Analysis:** Hypothesis Testing, Regression Modeling, Predictive Analytics
- **Tools & Platforms:** Jupyter Notebook, Git, VS Code, RStudio, Azure (basic), Office 365
- **Scripting & Automation:** Python, Bash, Process Automation
- **Collaboration & Communication:** Stakeholder Collaboration, Team Coordination, Analytical Reporting

Certification

- **SQL Intermediate Certificate**
  HackerRank
  Issued - May, 2025
- **Fundamentals of Data Governance**
  Edureka
  Issued - October, 2025
- **Healthcare Data Security, Privacy, and Compliance**
  Johns Hopkins University
  Issued - October, 2025
- **Process Data from Dirty to Clean**
  Google
  Issued - November, 2025
- **Supervised Machine Learning: Regression and Classification**
  Standford CPD
  Issued - August, 2024

Projects

- **Website Traffic & User Behavior Analysis**
  Technologies: SQL, Python (Pandas, NumPy), Google Analytics, Power BI, Tableau
  - Analyzed website traffic and user engagement data for multiple clients, helping them improve user experience and conversion rates.
  - Automated data extraction and reporting using Python and SQL, reducing manual effort and improving reporting efficiency.
  - Developed interactive dashboards in Power BI and Tableau to visualize user behavior trends and key performance indicators.
  - Conducted A/B testing on landing pages and UI elements, leading to a 20% increase in user retention for a key client.
- **Predicting Customer Sentiment in Social Media Interactions**
  Technologies: Python, NLP, Machine Learning (SVM, KNN, Bagging with RepTree), Sentiment Analysis, Data Preprocessing
  - Analyzed customer interactions on Amazon Help's Twitter account to understand sentiment trends.
  - Trained machine learning models to predict shifts in customer sentiment.
  - Identified K-Nearest Neighbor (KNN) and SVM as the most effective models for balancing accuracy and performance.
  - Showcased how sentiment analysis can improve customer support and engagement strategies.
- **Early Alzheimer's Disease Detection Using Machine Learning**
  Technologies: Python, Random Forest, Logistic Regression, CNN, Data Processing, MRI Analysis
  - Developed machine learning models to detect early signs of Alzheimer's Disease using MRI data.
  - Processed and analyzed the OASIS MRI dataset to identify key imaging patterns.
  - Achieved the highest accuracy and AUC score with Convolutional Neural Networks (CNNs).
  - Highlighted how AI-driven early detection can improve disease management and intervention.
- **Lung Cancer Prognostication with Machine Learning**
  Technologies: Python, XGBoost, LightGBM, CatBoost, Feature Engineering, ADASYN, Optuna
  - Built a stacking ensemble model combining XGBoost, LightGBM, and CatBoost for lung cancer prediction.
  - Improved prediction accuracy by handling class imbalance using Adaptive Synthetic Sampling (ADASYN).
  - Applied feature selection techniques like RFECV and LASSO to enhance model efficiency.
  - Used Bayesian optimization with Optuna to fine-tune hyperparameters for better performance.
  - Validated the model with Stratified K-Fold Cross-Validation, ensuring robustness for clinical applications.

Education

- **Masters in Computer Science**, University of South Dakota *May 2025*
  - ACM Club Member
  - USD Computing Club
  - BSA Treasurer

Research

- **A Multimodal Analytical Approach to Alzheimer's Disease Diagnosis Using Machine Learning and Convolutional Neural Networks on MRI Datasets**
  Published by **IEEE**
  - Investigated early detection of Alzheimer's Disease using machine learning and deep learning techniques on the OASIS MRI dataset.
  - Employed models such as Random Forest, Logistic Regression, Extra Trees, and Convolutional Neural Networks (CNN), achieving the highest accuracy and AUC with CNN.
  - Demonstrated the importance of early diagnosis to enable timely interventions and improve disease management strategies.
- **An Ontological Framework for Lung Carcinoma Prognostication via Sophisticated Stacking and Synthetic Minority Oversampling Techniques**
  Published by **IEEE**
  - Developed a stacking ensemble model combining XGBoost, LightGBM, and CatBoost, achieving improved predictive accuracy for lung cancer detection.
  - Addressed class imbalance with Adaptive Synthetic Sampling (ADASYN) and applied feature selection techniques like RFECV and LASSO Regression to enhance model performance.
  - Optimized hyperparameters using Bayesian techniques with Optuna and validated robustness through Stratified K-Fold Cross-Validation, demonstrating clinical applicability.
- **Recondite Thyroid Pathology Prediction: Hermeneutic Integration of Neural and Machine Learning Architectures**
  Published by **IEEE**
  - Developed an ensemble model integrating CNN, TabNet, and machine learning algorithms like XGBoost, achieving 98.7
  - Addressed class imbalance using advanced preprocessing techniques and optimized performance through feature selection, hyperparameter tuning, and cross-validation.
  - Demonstrated the effectiveness of combining deep learning and ensemble approaches for early and accurate diagnosis of thyroid disorders, improving clinical decision-making.
- **Predicting Customer Sentiment in Social Media Interactions: Analyzing Amazon Help Twitter Conversations Using Machine Learning**
  Published by **International Journal of Advanced Science Computing and Engineering**
  - Analyzed customer interactions with Amazon's support account "@AmazonHelp" to predict changes in sentiment using various machine learning algorithms.
  - Employed models like K-nearest neighbor, SVM, and Bagging with RepTree, identifying K-nearest neighbor and SVM as optimal for balancing accuracy and F-measure.
  - Demonstrated the potential of sentiment analysis and machine learning for enhanced customer engagement strategies on social platforms.

References

- **KC Santosh**
  University of South Dakota
  Department Chair
  Email: kc.santosh@usd.edu
  Phone: (929) 264-1429
- **Andrew Oldroyd**
  Rock IT Fuel Design and Tech
  Founder
  Email: andrew@atmwebdesign.ca
- **Mike Robles**
  Magic Web Studios
  CEO
  Email: mrobles@mtgi.net
  Phone: (605) 677-3184
- **Md Mahfujul Islam**
  Fora Financial LLC
  Senior Software Engineer
  Email: Mahfujul.islam@forafinancial.com
  Phone: (929) 264-1429